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Working with the Windows Event Log

From a SOC Analyst or IT Administrator's perspective, monitoring, collecting, and categorizing events occurring on all machines across the network is an invaluable source of information for defenders proactively analyzing and protecting their network from suspicious activity. On the other hand, attackers can see this as an opportunity to gain insight into the target environment, disrupt the flow of information, and as a way to hide their tracks. As we will see in later modules, sometimes we can find juicy information such as credentials hiding in event logs as a target system that we compromise during a penetration test. Other times, enumerating event logs can help us understand the level of logging in the environment (are just the defaults in place, or has the target organization configured more granular logging?). In this section, we will discuss the following:

- What is the Windows Event Log?
 - What information does it log, and where does it store this information?
 - Interacting with the Event Log via the `wevtutil` command line utility
 - Interacting with the Event Log using PowerShell cmdlets
-

What is the Windows Event Log?

A clear understanding of event logging is crucial to success in infosec. To kickstart our journey into gaining a thorough understanding of the Windows Event Log, there are a few key concepts that we need to define before diving in. These concepts will become the base upon which everything else will be built. The first one that needs to be explained is an **event** definition. Simply put, an **event** is any action or occurrence that can be identified and classified by a system's hardware or software. **Events** can be generated or triggered through a variety of different ways including some of the following:

- User-Generated Events
 - Movement of a mouse, typing on a keyboard, other user-controlled peripherals, etc.
- Application Generated Events
 - Application updates, crashes, memory usage/consumption, etc.
- System Generated Events
 - System uptime, system updates, driver loading/unloading, user login, etc.

With so many events occurring at different intervals of time from various sources, how does a Windows system keep track of and categorize all of them? This is where our second key concept, known as **event logging** comes into play.

Event Logging as defined by Microsoft:

"...provides a standard, centralized way for applications (and the operating system) to record important software and hardware events."

This definition sums up the question quite nicely. However, let us attempt to break it down a bit. As we discussed beforehand, there are a lot of events that are being triggered or generated concurrently on a system. Each event will have its own source that provides the information and details behind the event in its own format. So how does it handle all of this information?

Windows attempts to resolve this issue by providing a standardized approach to recording, storing, and managing events and event



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processes section, however, it is essential to understand that the **Event Log** is a required windows service starting upon system initialization that runs in the context of another executable and not it's own.

Before we dig into querying the Event Log from cmd.exe and PowerShell we need to understand the possible types of events available to us, the elements of a log, and various other elements.

Event Log Categories and Types

The main four log categories include application, security, setup, and system. Another type of category also exists called **forwarded events**.

Log Category	Log Description
System Log	The system log contains events related to the Windows system and its components. A system-level event could be a service failing at startup.
Security Log	Self-explanatory; these include security-related events such as failed and successful logins, and file creation/deletion. These can be used to detect various types of attacks that we will cover in later modules.
Application Log	This stores events related to any software/application installed on the system. For example, if Slack has trouble starting it will be recorded in this log.
Setup Log	This log holds any events that are generated when the Windows operating system is installed. In a domain environment, events related to Active Directory will be recorded in this log on domain controller hosts.
Forwarded Events	Logs that are forwarded from other hosts within the same network.

Event Types

There are five types of events that can be logged on Windows systems:

Type of Event	Event Description
Error	Indicates a major problem, such as a service failing to load during startup, has occurred.
Warning	A less significant log but one that may indicate a possible problem in the future. One example is low disk space. A Warning event will be logged to note that a problem may occur down the road. A Warning event is typically when an application can recover from the event without losing functionality or data.
Information	Recorded upon the successful operation of an application, driver, or service, such as when a network driver loads successfully. Typically not every desktop application will log an event each time they start, as this could lead to a considerable amount of extra "noise" in the logs.
Success Audit	Recorded when an audited security access attempt is successful, such as when a user logs on to a system.
Failure Audit	Recorded when an audited security access attempt fails, such as when a user attempts to log in but types their password in wrong. Many audit failure events could indicate an attack, such as Password Spraying.

Event Severity Levels

Each log can have one of five severity levels associated with it, denoted by a number:



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Information	4	An event that occurred on the system but did not cause any issues.
Warning	3	A potential problem that a sysadmin should dig into.
Error	2	An issue related to the system or service that does not require immediate attention.
Critical	1	This indicates a significant issue related to an application or a system that requires urgent attention by a sysadmin that, if not addressed, could lead to system or application instability.

Elements of a Windows Event Log

The Windows Event Log provides information about hardware and software events on a Windows system. All event logs are stored in a standard format and include the following elements:

- **Log name:** As discussed above, the name of the event log where the events will be written. By default, events are logged for **system**, **security**, and **applications**.
- **Event date/time:** Date and time when the event occurred
- **Task Category:** The type of recorded event log
- **Event ID:** A unique identifier for sysadmins to identify a specific logged event
- **Source:** Where the log originated from, typically the name of a program or software application
- **Level:** Severity level of the event. This can be information, error, verbose, warning, critical
- **User:** Username of who logged onto the host when the event occurred
- **Computer:** Name of the computer where the event is logged

There are many Event IDs that an organization can monitor to detect various issues. In an Active Directory environment, [this list](#) includes key events that are recommended to be monitored for to look for signs of a compromise. [This](#) searchable database of Event IDs is worth perusing to understand the depth of logging possible on a Windows system.

Windows Event Log Technical Details

The Windows Event Log is handled by the **EventLog** services. On a Windows system, the service's display name is **Windows Event Log**, and it runs inside the service host process **svchost.exe**. It is set to start automatically at system boot by default. It is difficult to stop the EventLog service as it has multiple dependency services. If it is stopped, it will likely cause significant system instability. By default, Windows Event Logs are stored in **C:\Windows\System32\winevt\logs** with the file extension **.evtx**.

Working with the Windows Event Log

```
PS C:\htb> ls C:\Windows\System32\winevt\logs
```

```
Directory: C:\Windows\System32\winevt\logs
```

Mode	LastWriteTime	Length	Name
-a----	11/16/2022 2:19 PM	7409664	Application.evtx
-a----	6/14/2022 8:20 PM	69632	HardwareEvents.evtx
-a----	6/14/2022 8:20 PM	69632	Internet Explorer.evtx
-a----	6/14/2022 8:20 PM	69632	Key Management Service.evtx
-a----	8/23/2022 7:01 PM	69632	Microsoft-Client-License-Flexible-Platform%4Admin.evtx
-a----	11/16/2022 2:19 PM	1052672	Microsoft-Client-Licensing-Platform%4Admin.evtx



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We can interact with the Windows Event log using the [Windows Event Viewer](#) GUI application via the command line utility [wevtutil](#), or using the [Get-WinEvent](#) PowerShell cmdlet. Both [wevtutil](#) and [Get-WinEvent](#) can be used to query Event Logs on both local and remote Windows systems via cmd.exe or PowerShell.

Interacting with the Windows Event Log - wevtutil

The [wevtutil](#) command line utility can be used to retrieve information about event logs. It can also be used to export, archive, and clear logs, among other commands.

Wevtutil without Parameters

Working with the Windows Event Log

```
C:\htb> wevtutil /?
```

Windows Events Command Line Utility.

Enables you to retrieve information about event logs and publishers, install and uninstall event manifests, run queries, and export, archive, and clear logs.

Usage:

You can use either the short (for example, ep /uni) or long (for example, enum-publishers /unicode) version of the command and option names. Commands, options and option values are not case-sensitive.

Variables are noted in all upper-case.

```
wevtutil COMMAND [ARGUMENT [ARGUMENT] ...] [/OPTION:VALUE [/OPTION:VALUE] ...]
```

Commands:

el		enum-logs	List log names.
gl		get-log	Get log configuration information.
sl		set-log	Modify configuration of a log.
ep		enum-publishers	List event publishers.
gp		get-publisher	Get publisher configuration information.
im		install-manifest	Install event publishers and logs from manifest.
um		uninstall-manifest	Uninstall event publishers and logs from manifest.
qe		query-events	Query events from a log or log file.
gli		get-log-info	Get log status information.
epl		export-log	Export a log.
al		archive-log	Archive an exported log.
cl		clear-log	Clear a log.

<SNIP>

We can use the [el](#) parameter to enumerate the names of all logs present on a Windows system.

Enumerating Log Sources

Working with the Windows Event Log

```
C:\htb> wevtutil el
```

```
AMSI/Debug
AirSpaceChannel
Analytic
Application
```



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```
ForwardedEvents
General Logging
HardwareEvents
```

```
<SNIP>
```

With the `gl` parameter, we can display configuration information for a specific log, notably whether the log is enabled or not, the maximum size, permissions, and where the log is stored on the system.

Gathering Log Information

Working with the Windows Event Log

```
C:\htb> wevtutil gl "Windows PowerShell"

name: Windows PowerShell
enabled: true
type: Admin
owningPublisher:
isolation: Application
channelAccess: 0:BAG:SYD:(A;;;0x2;;;S-1-15-2-1)(A;;;0x2;;;S-1-15-3-1024-3153509613-960666767-3724611135-27256626-
logging:
  logFileName: %SystemRoot%\System32\Winevt\Logs\Windows PowerShell.evtx
  retention: false
  autoBackup: false
  maxSize: 15728640
publishing:
  fileMax: 1
```

The `gli` parameter will give us specific status information about the log or log file, such as the creation time, last access and write times, file size, number of log records, and more.

Working with the Windows Event Log

```
C:\htb> wevtutil gli "Windows PowerShell"

creationTime: 2020-10-06T16:57:38.617Z
lastAccessTime: 2022-10-26T19:05:21.533Z
lastWriteTime: 2022-10-26T19:05:21.533Z
fileSize: 11603968
attributes: 32
numberOfLogRecords: 9496
oldestRecordNumber: 1
```

There are many ways we can query for events. For example, let's say we want to display the last 5 most recent events from the Security log in text format. Local admin access is needed for this command.

Querying Events

Working with the Windows Event Log

```
C:\htb> wevtutil qe Security /c:5 /rd:true /f:text

Event[0]
  Log Name: Security
```



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Opcode: Info
Keyword: Audit Success
User: N/A
User Name: N/A
Computer: ICL-WIN11.greenhorn.corp
Description:
A security-enabled local group membership was enumerated.

Subject:
Security ID: S-1-5-18
Account Name: ICL-WIN11\$
Account Domain: GREENHORN
Logon ID: 0x3E7

Group:
Security ID: S-1-5-32-544
Group Name: Administrators
Group Domain: Builtin

Process Information:
Process ID: 0x56c
Process Name: C:\Windows\System32\svchost.exe

Event[1]
Log Name: Security
Source: Microsoft-Windows-Security-Auditing
Date: 2022-11-16T14:54:13.0160000Z
Event ID: 4672
Task: Special Logon
Level: Information
Opcode: Info
Keyword: Audit Success
User: N/A
User Name: N/A
Computer: ICL-WIN11.greenhorn.corp
Description:
Special privileges assigned to new logon.

Subject:
Security ID: S-1-5-21-4125911421-2584895310-3954972028-1001
Account Name: htb-student
Account Domain: ICL-WIN11
Logon ID: 0x8F211

Privileges:
SeSecurityPrivilege
SeTakeOwnershipPrivilege
SeLoadDriverPrivilege
SeBackupPrivilege
SeRestorePrivilege
SeDebugPrivilege
SeSystemEnvironmentPrivilege
SeImpersonatePrivilege
SeDelegateSessionUserImpersonatePrivilege

Event[2]
Log Name: Security
Source: Microsoft-Windows-Security-Auditing
Date: 2022-11-16T14:54:13.0160000Z
Event ID: 4624
Task: Logon
Level: Information
Opcode: Info
Keyword: Audit Success
User: N/A
User Name: N/A
Computer: ICL-WIN11.greenhorn.corp
Description:
An account was successfully logged on.



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Logon ID: 0x3E7

<SNIP>

We can also export events from a specific log for offline processing. Local admin is also needed to perform this export.

Exporting Events

Working with the Windows Event Log

```
C:\htb> wevtutil epl System C:\system_export.evtx
```

Interacting with the Windows Event Log - PowerShell

Similarly, we can interact with Windows Event Logs using the [Get-WinEvent](#) PowerShell cmdlet. Like with the [wevtutil](#) examples, some commands require local admin-level access.

To start, we can list all logs on the computer, giving us the number of records in each log.

PowerShell - Listing All Logs

Working with the Windows Event Log

```
PS C:\htb> Get-WinEvent -ListLog *
```

LogMode	MaximumSizeInBytes	RecordCount	LogName
Circular	15728640	657	Windows PowerShell
Circular	20971520	10713	System
Circular	20971520	26060	Security
Circular	20971520	0	Key Management Service
Circular	1052672	0	Internet Explorer
Circular	20971520	0	HardwareEvents
Circular	20971520	6202	Application
Circular	1052672		Windows Networking Vpn Plugin Platform/Op...
Circular	1052672		Windows Networking Vpn Plugin Platform/Op...
Circular	1052672	0	SMSApi
Circular	1052672	61	Setup
Circular	15728640	24	PowerShellCore/Operational
Circular	1052672	99	OpenSSH/Operational
Circular	1052672	46	OpenSSH/Admin

<SNIP>

We can also list information about a specific log. Here we can see the size of the [Security](#) log.

Security Log Details

Working with the Windows Event Log

```
PS C:\htb> Get-WinEvent -ListLog Security
```

LogMode	MaximumSizeInBytes	RecordCount	LogName
Circular	20971520	26060	Security



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list the last five events recorded in the Security log. By default, the newest logs are listed first. If we want to get older logs first, we can reverse the order to list the oldest ones first using the `-Oldest` parameter.

Querying Last Five Events

Working with the Windows Event Log

```
PS C:\htb> Get-WinEvent -LogName 'Security' -MaxEvents 5 | Select-Object -ExpandProperty Message

An account was logged off.

Subject:
  Security ID:      S-1-5-111-3847866527-469524349-687026318-516638107-1125189541-6052
  Account Name:     sshd_6052
  Account Domain:   VIRTUAL USERS
  Logon ID:         0x8E787

Logon Type:        5

This event is generated when a logon session is destroyed. It may be positively correlated with a logon event or
Special privileges assigned to new logon.

Subject:
  Security ID:      S-1-5-18
  Account Name:     SYSTEM
  Account Domain:   NT AUTHORITY
  Logon ID:         0x3E7

Privileges:        SeAssignPrimaryTokenPrivilege
                  SeTcbPrivilege
                  SeSecurityPrivilege
                  SeTakeOwnershipPrivilege
                  SeLoadDriverPrivilege
                  SeBackupPrivilege
                  SeRestorePrivilege
                  SeDebugPrivilege
                  SeAuditPrivilege
                  SeSystemEnvironmentPrivilege
                  SeImpersonatePrivilege
                  SeDelegateSessionUserImpersonatePrivilege

An account was successfully logged on.

<SNIP>
```

We can dig deeper and look at specific event IDs in specific logs. Let's say we only want to look at logon failures in the Security log, checking for Event ID [4625: An account failed to log on](#). From here, we could use the `-ExpandProperty` parameter to dig deeper into specific events, list logs from oldest to newest, etc.

Filtering for Logon Failures

Working with the Windows Event Log

```
PS C:\htb> Get-WinEvent -FilterHashTable @{LogName='Security';ID='4625 '}

ProviderName: Microsoft-Windows-Security-Auditing

TimeCreated          Id LevelDisplayName Message
-----
11/16/2022 2:53:16 PM 4625 Information An account failed to log on....
11/16/2022 2:53:16 PM 4625 Information An account failed to log on....
11/16/2022 2:53:12 PM 4625 Information An account failed to log on....
11/16/2022 2:50:36 PM 4625 Information An account failed to log on
```




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We can also look at only events with a specific information level. Let's check all System logs for only **critical** events with information level **1**. Here we see just one log entry where the system did not reboot cleanly.

Working with the Windows Event Log

```
PS C:\htb> Get-WinEvent -FilterHashTable @{LogName='System';Level='1'} | select-object -ExpandProperty Message
```

The system has rebooted without cleanly shutting down first. This error could be caused if the system stopped

Practice more with **wevtutil** and **Get-WinEvent** to become more comfortable with searching logs. Microsoft provides some [examples](#) for **Get-WinEvent**, while [this site](#) shows examples for **wevtutil**, and [this site](#) has some additional examples for using **Get-WinEvent**.

Moving On

This section introduced the Windows Event Log, a vast topic that we will dig much deeper into in later modules. Try out the various examples in this section and get comfortable using both tools to query for specific information. In later modules, we will see how we can sometimes find sensitive data, such as passwords, in Event Logs. Logging on Windows is very powerful when configured properly. Each system generates a massive amount of logs, and, as we saw with all the possible Event IDs, we can get quite granular with what exactly we choose to log. All of this data on its own would be very difficult to constantly query and is most effective when forwarded to a SIEM tool that can be used to set up alerts on specific Event IDs which may be indicative of an attack, such as Kerberoasting, Password Spraying, or other less common attacks. As penetration testers, we should be familiar with Windows Event Log, how we can use it to gain information about the environment, and sometimes even extract sensitive data. For blue teamers, in-depth knowledge of Windows Event Log and how to leverage it for effective alerting and monitoring is critical.

In the next section, we will cover working with networking operations from the command line on a Windows system.

VPN Servers

Warning: Each time you "Switch", your connection keys are regenerated and you must re-download your VPN connection file.

All VM instances associated with the old VPN Server will be terminated when switching to a new VPN server.

Existing PwnBox instances will automatically switch to the new VPN server.

PROTOCOL

● UDP 1337 ● TCP 443

DOWNLOAD VPN CONNECTION FILE



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∞ / 1 spawns left

Waiting to start...



Questions

Answer the question(s) below to complete this Section and earn cubes!

Target(s): Fetching status...

SSH to with user "htb-student" and password "HTB_@cademy_stdnt!"

+ 0 Explore the targets provided and practice your Event Log PowerShell Kung-Fu. Type COMPLETE as the answer when finished.

Submit your answer here...

+10 Streak pts

Submit

Hint

Show Solution

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Gathering System Information





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Finish Strong

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Beyond This Module

My Workstation

OFFLINE

 Start Instance

 / 1 spawns left